# Abirami Kumar

(510) 309-9098 | abirami.kumar@berkeley.edu | linkedin.com/in/-abiramikumar | abirami07.github.io

## EDUCATION

**University of California, Berkeley** **Berkeley, CA**
*B.A. Data Science & B.A. Computer Science* *Aug 2025 – May 2028*

- **Relevant Coursework:** Data Structures & Algorithms, Principles and Techniques of Data Science, Linear Algebra & Differential Equations, Discrete Mathematics, Computational Structures in Data Science
- **Student Organizations:** Nova Consulting, Software Product, Data Discovery Program, CS Kickstart

## EXPERIENCE

**Software Engineering Intern** **Remote**
*AP0110* *May 2026 – Aug 2026*

- Built a natural-language extraction pipeline for a React-based AI assistant, parsing user inputs, normalizing units, validating physiological ranges, and detecting symptom negation to support reliable user interactions.
- Engineered a JavaScript rules engine evaluating 6 physiological metrics, implementing deterministic risk classification, missing-data handling, and symptom-aware validation across edge cases.
- Developed a Python/Pandas ETL pipeline integrating 3 NHANES datasets into 4,907 validated adult records, implementing data cleaning, risk classification, and structured data exports.

**AI Researcher** **Berkeley, CA**
*KnaiTai Open Initiative* *August 2026 – Present*

- Developed Python-based knowledge activation and runtime-selection logic within the A4 architecture, enabling context-aware evidence retrieval and dynamic AI reasoning.
- Contributed to automated regression and integration testing using Pytest across a suite of 57 passing tests, validating runtime operations, evidence freshness, and authorization decisions.
- Debugged stale evidence propagation in multi-cycle reasoning workflows, implementing state synchronization fixes and validating system behavior through end-to-end execution traces.

**Technical Analyst** **Remote**
*Cisco | Nova Consulting* *September 2026 – Present*

- Developing a structured competitive benchmarking framework to evaluate channel-partner profitability across enterprise technology vendors, analyzing margin structures, rebates, incentives, and service revenue models.
- Analyzing vendor documentation, financial disclosures, and partner-program data to identify competitive trends and translate findings into actionable recommendations for enterprise stakeholders.

## PROJECTS

**Lexia | UC Berkeley AI Hackathon | GitHub** **June 2026**

- Built and demoed a full-stack AI accessibility platform in 48 hours using React, FastAPI, and Python, dynamically adapting web content based on real-time reading behavior.
- Engineered AI workflows integrating Browserbase, Stagehand, Claude, and Deepgram for browser automation, content extraction, LLM-based comprehension, and voice interaction.
- Implemented gaze-based adaptation using PyGaze, analyzing fixation duration, regressions, and line-skipping signals to trigger personalized content transformations.

**Sather Sense: Indoor Navigation System | 1st Place at SodaHacks | GitHub** **March 2026**

- Built a cross-platform indoor navigation application using React Native, TypeScript, and Expo, delivering real-time voice guidance and haptic navigation for visually impaired users.
- Developed a React/Vite administration interface for interactive floor-plan annotation and navigation graph management, enabling users to configure multi-floor indoor navigation workflows.
- Implemented graph-based pathfinding across multi-floor building layouts and OCR-assisted localization for indoor positioning.
- Developed frontend tests using React Testing Library to validate component behavior and user interactions, improving reliability across navigation workflows.

## SKILLS

**Languages:** Python, Java, SQL, TypeScript, JavaScript, HTML, CSS
**Frameworks & Libraries:** React, React Native, Node.js, FastAPI, Expo, Vite, React Testing Library, Pandas, NumPy
**Developer Tools:** Git, GitHub, Linux, Pytest, REST APIs